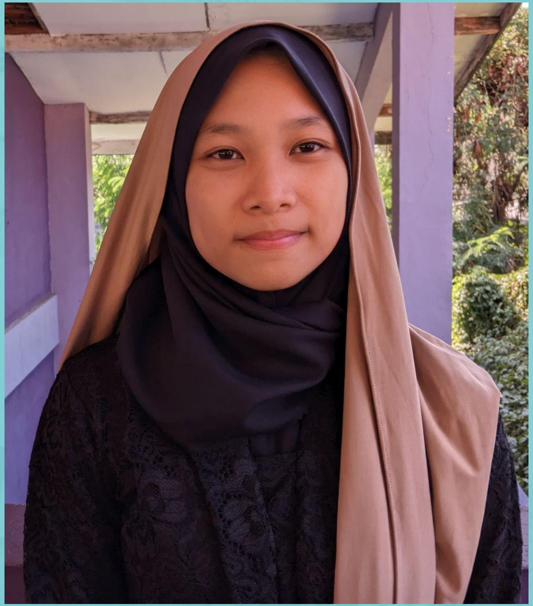
Rakamin Academy X kimia farma

Analisis Kinerja Bisnis Kimia Farma Tahun 2020-2023

Kimia Farma - Big Data Analytics

Presented by

Dian Ayu Rahmawati



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Dian Ayu Rahmawati

Halo! Aku Dian Ayu Rahmawati, biasa dipanggil 'Dian'. Aku, lulusan SMKN 12 Surabaya jurusan Pengembangan Perangkat Lunak dan Gim.

Aku suka dunia data karena seru aja lihat gimana angka bisa bantu ngambil keputusan yang real buat bisnis. Sekarang aku lagi aktif ikut program, bootcamp, dan buat berbagai project

Sawotratap, Gedangan, Sidoarjo, Jawa Timur

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Courses and Certification

DQLab Study Case Bootcamp Data Analyst with Excel | Link Sertifikat	05 Oktober 2025
DQLab Mini Bootcamp: Introduction to Data Analyst | Link Sertifikat	Oktober 2025
MySkill Short Class: Data Visualization with Looker Data Studio | Link Sertifikat	11 September 2025
RevoU Mini Course: Intro to Data Analyst | Link Sertifikat	22 Agustus 2025
EF SET Certificate | Link Sertifikat	1 Agustus 2025
Digitalent Intermediate Junior Web Developer | Link Sertifikat	September 2025

About Company

Industri: Farmasi & Retail Apotek

Layanan: Penjualan obat, farmasi, pelayanan kesehatan

Kimia Farma adalah salah satu perusahaan farmasi terbesar di Indonesia, dan pastinya kita sering lihat apoteknya di mana-mana. Di project ini, aku menganalisis performa penjualan dan layanan cabang Kimia Farma selama 2020–2023. Targetnya adalah memahami tren bisnis dan insight pelanggan dari data realistik skala enterprise.

Project Portfolio

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Analisis performa penjualan Kimia Farma (2020–2023)

Dataset: transaksi, produk, cabang, inventory

Output:

✅ Tabel analisa di BigQuery

✅ SQL Query Insights

✅ Dashboard Google Looker Studio - Dashboard Dian Ayu

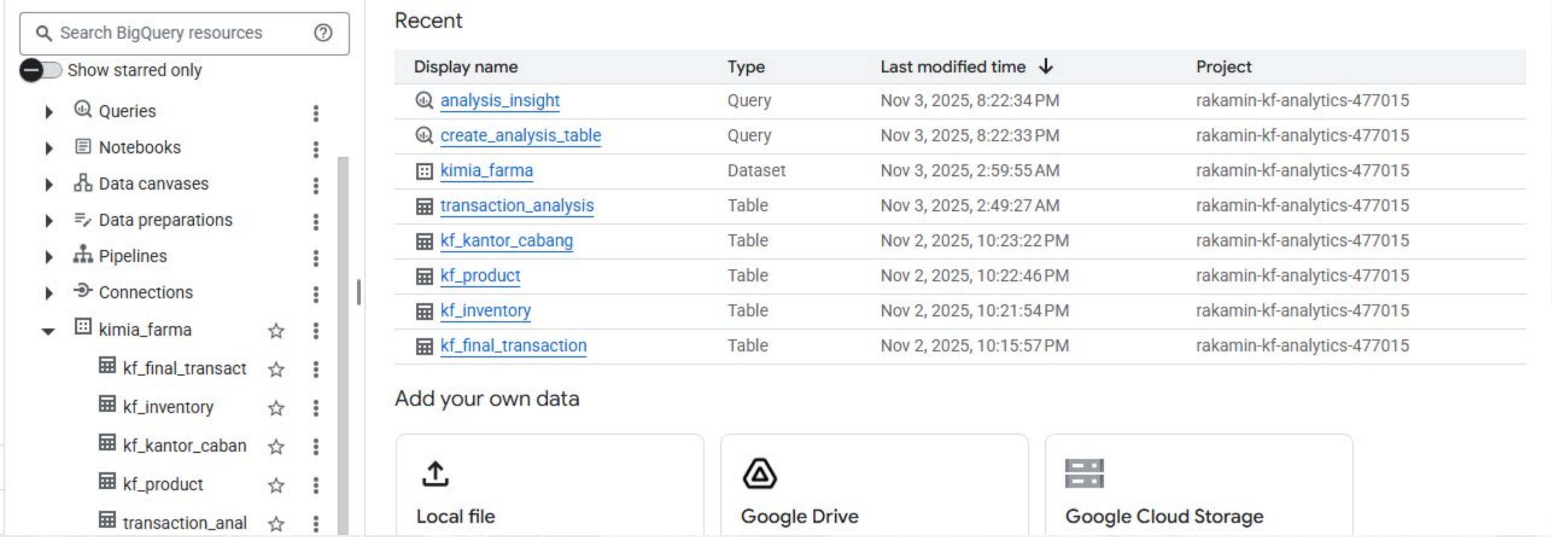
✅ GitHub Repo - Repo Link

✅ Video Presentasi - Link Video

Project ini fokus buat nge-explore performa bisnis Kimia Farma dari sisi sales, profit, dan rating customer per cabang. Data yang disediakan mencakup transaksi, info produk, data cabang, dan inventory. Setelah itu, aku bikin data mart di BigQuery, lanjut visualisasi insightnya ke Looker Studio supaya lebih gampang dianalisis.

1. Importing Dataset to BigQuery

1. **Pertama, semua dataset aku upload ke BigQuery dalam satu dataset bernama kimia_farma.**
2. **Setelah upload semua tabel, aku cek struktur, sample data, dan count-nya.**
3. **Baru transaksi di-join jadi satu tabel analisa dengan SQL.**



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2. Tabel Analisa

- **Tabel: kimia_farma.transaction_analysis**
- **Gabung data transaksi + produk + cabang**
- **Hitung:**
  1. **Nett Sales**
  2. **Persentase Laba**
  3. **Nett Profit**

Ini tabel inti buat analisisnya. Intinya aku join empat tabel untuk bikin satu data mart lengkap. Di sini aku juga hitung nett sales dan nett profit berdasarkan ketentuan margin harga. Jadi final table-nya siap dipakai buat dashboard dan query insight lain.

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Schema | Details | Preview | Table Explorer Preview

Field name	Type	Mode
transaction_id	STRING	NULLABLE
date	DATE	NULLABLE
branch_id	INTEGER	NULLABLE
branch_name	STRING	NULLABLE
kota	STRING	NULLABLE
provinsi	STRING	NULLABLE
rating_cabang	FLOAT	NULLABLE
customer_name	STRING	NULLABLE
product_id	STRING	NULLABLE
product_name	STRING	NULLABLE
actual_price	INTEGER	NULLABLE
discount_percentage	FLOAT	NULLABLE
persentase_gross_laba	FLOAT	NULLABLE
nett_sales	FLOAT	NULLABLE
nett_profit	FLOAT	NULLABLE
rating_transaksi	FLOAT	NULLABLE

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3. BigQuery Syntax

SQL aku bagi dua — satu khusus buat bikin data mart, dan satu lagi buat query insight yang nanti dipakai juga buat dashboard. Query-nya mencakup tren penjualan per tahun, ranking cabang per provinsi, sampai ngecek gap antara rating cabang dan rating transaksi.

File SQL:

create_analysis_table.sql

analysis_insight_queries.sql

```sql
-- Year Over Year Nett Sales
-- ============================
SELECT
  EXTRACT(YEAR FROM date) AS year,
  SUM(nett_sales) AS total_sales
FROM kimia_farma.transaction_analysis
GROUP BY year
ORDER BY year;

-- Top 10 Total Transaksi per Provinsi
-- ============================
SELECT
  provinsi,
  COUNT(transaction_id) AS total_transactions
FROM kimia_farma.transaction_analysis
GROUP BY provinsi
ORDER BY total_transactions DESC
LIMIT 10;

-- Top 10 Nett Sales per Provinsi
-- ============================
SELECT
  provinsi,
  SUM(nett_sales) AS total_nett_sales
FROM kimia_farma.transaction_analysis
GROUP BY provinsi
ORDER BY total_nett_sales DESC
LIMIT 10;

-- Top 5 Branch High Rating Cabang but Low Rating Transaksi
-- ============================
SELECT
  branch_name,
  rating_cabang,
  AVG(rating_transaksi) AS avg_rating_transaksi
FROM kimia_farma.transaction_analysis
GROUP BY branch_name, rating_cabang
ORDER BY rating_cabang DESC, avg_rating_transaksi ASC
LIMIT 5;

-- Total Profit per Province (Geo Map)
-- ============================
SELECT
  provinsi,
  SUM(nett_profit) AS total_profit
FROM kimia_farma.transaction_analysis
GROUP BY provinsi
ORDER BY total_profit DESC;
```

```sql
CREATE OR REPLACE TABLE kimia_farma.transaction_analysis AS
SELECT
   t.transaction_id,
   t.date,
   c.branch_id,
   c.branch_name,
   c.kota,
   c.provinsi,
   c.rating AS rating_cabang,
   t.customer_name,
   p.product_id,
   p.product_name,
   t.price AS actual_price,
   t.discount_percentage,
   CASE
       WHEN t.price <= 50000 THEN 0.10
       WHEN t.price <= 100000 THEN 0.15
       WHEN t.price <= 300000 THEN 0.20
       WHEN t.price <= 500000 THEN 0.25
       ELSE 0.30
   END AS persentase_gross_laba,
   t.price * (1 - t.discount_percentage) AS nett_sales,
   (t.price * (1 - t.discount_percentage)) *
       CASE
           WHEN t.price <= 50000 THEN 0.10
           WHEN t.price <= 100000 THEN 0.15
           WHEN t.price <= 300000 THEN 0.20
           WHEN t.price <= 500000 THEN 0.25
           ELSE 0.30
       END AS nett_profit,
   t.rating AS rating_transaksi
FROM `kimia_farma.kf_final_transaction` t
LEFT JOIN `kimia_farma.kf_product` p USING(product_id)
LEFT JOIN `kimia_farma.kf_kantor_cabang` c USING(branch_id);
```

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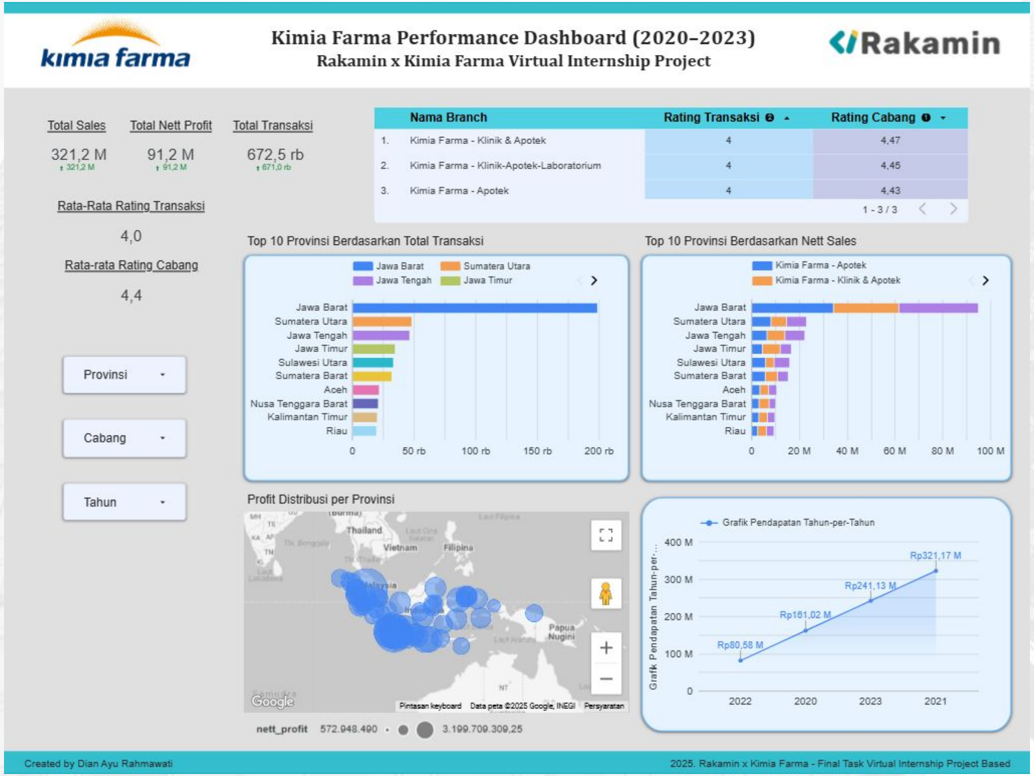
4. Dashboard Performance Analytics

Setelah datanya siap, aku bikin dashboard interaktif di Looker Studio. Ada KPI utama, grafik YoY, ranking provinsi, sampai geo-map profit biar keliatan performa regional-nya.

[Dashboard Link] Ini isi dashboardnya:



✅ **KPI scorecard (Sales, Profit, Rating)**

✅ **Time series trend**

✅ **Top 10 cabang**

✅ **Geo map profit**

✅ **Filter interaktif**

Thank You

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